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Classifying Supernova Transient 2026atw

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ABSTRACT

In this paper, I set to classify the previously unclassified supernova transient, SN 2026atw, for my Astronomy 100 final project. Visible from the Fred Lawrence Whipple Observatory in Arizona, the FAST (Fast Spectrograph for the Tillinghast Telescope) spectrometer was used to observe the transient on the night of March 17th, 2026. All of the raw data, in the form of .fits files, were reduced and used to create a spectrum of SN 2026atw, where the presence of Balmer lines immediately classified it as Type II and determined its progenitor as a massive main sequence star turned red supergiant that underwent core collapse. Further measurements on the redshift of the spectrum and calculations were performed, such as the ejecta velocity, and comparing this to the recession velocity and distance to the host galaxy of SN 2026atw. Then, the light curve of SN 2026atw from ATLAS forced photometry was obtained and plotted in absolute magnitude using the redshift of the supernova, where the peak magnitude, phase, and subtype "P" were determined. The measurements and calculations made by me for SN 2026atw were then compared across literature, and then compared to SNID-SAGE's classification of SN 2026atw, which confirmed that it was a Type IIP supernova. SN 2026atw's classification was then published on the Transient Name Server for 2026atw; but further modeling work on the light curve remains to determine the mass and size of the progenitor, as well as the total energy of the supernova explosion.

1. INTRODUCTION

Transient astronomy is the observation of astrophysical phenomena that change their brightness, or flux, over a period of time (Astrobites Collaboration 2022). One such observable transient are supernovae explosions, which change in brightness over the course of the explosion. The precursor to a supernova transient is called the progenitor, which is a label used to classify what astronomical object existed before the explosion. Progenitors are primarily either massive stars that undergo core-collapse or white dwarfs that undergo thermonuclear fusion once they reach the Chandrasekhar limit at $\approx 1.44\,M_\odot$ (NASA Space Place 2021). Determining the progenitor is at the core of the purpose in observing and classifying supernova transients, as one can determine the progenitor and its physical properties from spectral and light curve analysis on any observed supernova transient. Doing so not only allows us to better understand the stellar lifetimes of massive stars or white dwarfs and derive properties of the transient host galaxies, but to also use them as standard candles (in the case of supernovae with white dwarf progenitors) that allow us to accurately measure distances to galaxies and observe the accelerated expansion of the universe called dark energy (NASA Science 2024).

There are two main, standard classifications for supernovae: Type I and Type II. The difference between classifying a supernova as Type I or Type II is based on the presence of any hydrogen lines in its spectra; if hydrogen lines are absent, it is classified as Type I, and if they are present, it is classified as Type II (The Schools' Observatory n.d.).

For Type I supernovae, the subtypes are Type Ia, Type Ib, and Type Ic. Type Ia supernovae have white dwarf progenitors, while Types Ib and Ic have massive star progenitors. White dwarfs are the cores of dead stars that were between 0.08 - 8 $M_\odot$ (OpenSpace Project 2026) in their main sequence lifetime before they became red giants. As these red giants die, they shed off their layers in a planetary nebula, revealing the white dwarf core composed primarily of helium, carbon, and oxygen once all layers have been shed. White dwarfs achieve thermonuclear fusion, resulting in a thermonuclear or Type Ia supernova, at the Chandrasekhar limit of $\approx 1.44\,M_\odot$ when they are in binary system with another main sequence star and the gravitational force of the white dwarf attracts the other star's material onto itself and becomes more massive. Once the Chandrasekhar limit is reached, the white dwarf explodes as a Type Ia supernova (The Schools' Observatory n.d.). Because the white dwarf is not composed of nearly any hydrogen at all, this explains the absence of hydrogen lines in Type Ia supernova spectra.

Type Ib and Type Ic supernovae have massive star progenitors $> 8M_{\odot}$, but have become stripped of their hydrogen layers once their cores collapse. Given the absence of these hydrogen layers in the core collapse of the massive stars, there is no hydrogen present in the resulting supernova spectrum, and Type Ib and Type Ic supernovae are deemed stripped-envelope core-collapse supernovae (The Schools' Observatory n.d.).

For Type II supernovae, the subtypes are Type IIL, Type IIP, Type IIn, and Type IIb. The progenitors of these supernovae are all from main sequence stars of a mass $> 8M_{\odot}$, but they still have their hydrogen layers during core-collapse. The difference in subtyping for Type II supernovae comes from analyzing the shapes of their light curves and spectra, where "L" stands for linear light curve, "P" stands for plateau light curve (Swinburne University of Technology 2026a), "n" stands for narrow hydrogen lines, and "b" stands for broad hydrogen lines in spectra (Andrews 2011).

Given that observing and classifying a supernova transient allows us determine its progenitor and its physical properties, as well as the characteristic properties of the supernova explosion itself and the host galaxy, I set out to see if I could classify a supernova transient myself to do as such. The supernova that I chose was 2026atw, an initially unclassified supernova in the Transient Name Server (Transient Name Server 2026) that was observable from the Fred Lawrence Whipple Observatory (FLWO) in Arizona during my Astronomy 100 spring break class trip. I set out to observe supernova (SN) 2026atw using the FAST (Fast Spectrograph for the Tillinghast Telescope) spectrometer so that I would not only obtain a raw spectra of the SN 2026atw transient, but also data reduce it, calibrate it using a standard star observed in the same night, and determine the following: the supernova type, progenitor, phase (age of supernova), peak magnitude, redshift of SN 2026atw, and velocity of ejecta for SN 2026atw. I also set out to determine the redshift, velocity of and distance to SN 2026atw's host galaxy. After spectral and light curve analysis, I was able to classify SN 2026atw as a Type IIP supernova with a massive main sequence star $> 8M_{\odot}$ progenitor, which was confirmed using the SNID-SAGE supernova classifying software and comparing SN 2026atw to both other Type IIP supernova in literature, as well as comparing to a Type Ia supernova to emphasize the differences in classification.

2. DATA

2.1. *Choosing SN 2026atw*

SN 2026atw was observed using the FAST Spectrometer at the FLWO in Arizona, which can observe a Dec range of roughly -30º to +80º declination (Smithsonian Astrophysical Observatory 2026). The RA and Dec coordinates of SN 2026atw from the Transient Name Server are 262.250333 RA and +56.780078 Dec (Transient Name Server 2026), which was observable from FLWO. At the time of observation, the night of March 17th, 2026, SN 2026atw was still unclassified, but observed with light curve data from the ATLAS, LAST, GOTO, and WFST reporting groups. Given that I wanted to observe and classify a transient that had available light curve data, I chose SN 2026atw from the various supernova transient observations made the night of March 17th. The specific raw data file I used for the SN 2026atw is titled 00080.2026atw.fits.

2.2. *Observation Files Specifics*

On the same night, 41 BIAS and 40 FLAT fits files were also taken, which would be used in the data reduction of the raw SN 2026atw spectrum to generate the uncalibrated science spectrum. The Ar lamp that was used for reference spectral lines was 0080.COMP.fits, which was taken with SN 2026atw and used to produce the uncalibrated flux spectrum in ADU flux units. The standard star observed the same night is Hiltner 600, which has known physical fluxes that were used to calibrate the observed Hilter 600 spectrum to derive a sensitivity function to calibrate the uncalibrated flux SN 2026atw spectrum. The raw data file I used for the star is 0053.Hiltner600.fits, and the Ar lamp used was 0054.COMP.fits. The known physical fluxes of Hiltner 600 came from the fhilt600.dat.txt file, which could be found online at European South Observatory database (European Southern Observatory 2026).

2.3. *Data Reduction of Observed Targets*

In order to reduce the raw data image from 0080.2026atw.fits and 0053.Hiltner600.fits into scientifically usable spectra, I created an .ipynb notebook in Google Colab, where I imported the raw FAST spectroscopic data as FITS files. Using the BIAS frames collected from the night of observation, they were overscan-corrected using the median signal in the overscan region, then trimmed to the illuminated CCD before being combined into a median master BIAS frame. The median master BIAS frame removes the detector readout signature from the raw spectra. The flat-field frames (FLAT frames) were processed in the same manner before being median-combined and normalized by their median value, which created a master FLAT that corrected for pixel-to-pixel sensitivity variations in both the SN 2026atw and Hiltner 600 images. After, the science exposure of these spectra were bias-subtracted via the master BIAS before being divided by the normalized master FLAT to produce the reduced two-dimensional spectra.

By collapsing the two-dimensional spectra along the dispersion direction, a spatial profile of the images were computed. The peak of the spatial profiles were then used to identify the trace position of SN 206atw and Hiltner 600 on the CCD, and I could create one-dimensional spectra by extracting the summed flux within a small aperture centered at these trace peaks. By sampling the background regions above and below the trace, the sky subtraction of telluric lines was performed. This meant constructing a median sky spectrum, scaling it to the aperture width, and subtracting it from the spectra before applying a small median filter to suppress high-frequency noise in the spectra. Completing this step reduced the very prominent sky lines that are not useful for spectral analysis.

After reducing the sky lines, wavelength calibration was carried out using comparison lamp exposures (0080.COMP.fits for SN 2026atw, 0054.COMP.fits for 0053.Hiltner600.fits), which contains known Argon emission lines. These COMP files underwent overscan correction, trimming, and bias subtraction as well before one-dimensional arc spectra were also extracted using the same trace aperture as the science spectra. With the Ar lamp, the emission-line peaks in my spectra were identified and matched to the reference Argon wavelengths before fitting a second-order polynomial solution to convert pixel positions into wavelengths in Angstroms (Å). This produced uncalibrated flux spectra in flux ADU units for both 2026atw and Hiltner 600

In order to calibrate the uncalibrated flux spectra into real flux units, the Hiltner 600 spectrum was first converted from detector counts to count rate by using the exposure time:

$$R_{\rm std}(\lambda) = \frac{C_{\rm std}(\lambda)}{t_{\rm exp,std}} \tag{1}$$

where $C_{\rm std}(\lambda)$ is the extracted spectrum of Hiltner 600 in ADU and $t_{\rm exp,std}$ is the exposure time of the standard-star observation.

Then, the sensitivity function of the instrument was then derived by comparing $R_{\rm std}(\lambda)$ to the physical flux of Hiltner 600 found in fhilt600.dat.txt, $F_{\rm ref}(\lambda)$, in units of erg s^{-1} cm^{-2} Å^{-1}, which flux-calibrated the Hiltner 600 spectrum:

$$S(\lambda) = \frac{F_{\rm ref}(\lambda)}{R_{\rm std}(\lambda)} \tag{2}$$

The sensitivity function was then smoothed using a Gaussian smoothing to reduce high-frequency noise:

$$S_{\rm smooth}(\lambda) = \mathrm{GaussianSmooth}\left[S(\lambda)\right] \tag{3}$$

SN 2026atw's extracted spectrum was also converted into counts per second using the same formula:

$$R_{\rm SN}(\lambda) = \frac{C_{\rm SN}(\lambda)}{t_{\rm exp,SN}} \tag{4}$$

then finally, the flux-calibrated spectrum of SN 2026atw was produced by applying the sensitivity function to the counts of the SN spectrum:

$$F_{\rm SN}(\lambda) = R_{\rm SN}(\lambda) \times S_{\rm smooth}(\lambda) \tag{5}$$

where $F_{\rm SN}(\lambda)$ is the calibrated SN flux spectrum in physical units of erg s^{-1} cm^{-2} Å^{-1}. All of the files and code used in data reduction can be found in Section 7.

2.4. *Data Used for Figures*

Figure 4, an image of SN 2026atw's host galaxy, comes from the DESI Legacy Survey (DESI Legacy Imaging Surveys 2026) at the same RA and Dec coordinates of 262.250333 RA and +56.780078 Dec. Figure 5, a modified velocity vs days from explosion plot, comes from Figure 18 of Gutiérrez et al. (2017). The content of the data of Figure 18 were not changed in any way, but the figure itself was cropped as to utilize the necessary graph relevant to this project, which was for the Hα line.

The light curve data used to generate Figures 5 of SN 2026atw's light curve came from ATLAS Forced Photometry (ATLAS Forced Photometry Service 2026a). The data file itself is in apparent magnitudes, which were converted into absolute magnitudes given that I calculated the redshift of SN 2026atw using observed and rest Hα lines from my calibrated SN 2026atw spectrum.

In Figure 6, the light curve comparison of SN 2026atw, a Type IIP supernova, to a Type Ia supernova, I used SN 2017hjw at RA 77.182625 and Dec +70.4757, which had an excellent light curve to compare to and show the differences between Type IIP to and Type Ia supernovae (Tonry et al. 2018, Figure 12 on pg 17). The data file for SN 2017hjw can be found on ATLAS Forced Photometry

For the light curve comparison in Figure 7 of SN 2026atw to another Type II supernova, I used SN 2023ixf at RA 210.910674637 and Dec +54.3116510708RA from the Transient Name Server (Transient Name Server 2023). I felt that it would be useful to compare 2026atw to another Type II supernova in order to analyze similarities in their light curves and futher verify my classification. The data file for the light curve of SN 202ixf can be found on ATLAS Forced Photometry.

3. METHODOLOGY

3.1. *Spectral Analysis: Identifying Hydrogen Lines*

As previously mentioned, the first step one can take in classifying any supernova transient spectra as Type I or Type II and its progenitor is by determining if hydrogen is present in the spectrum. The two most prominent hydrogen lines are featured in the Balmer lines, which include the very prominent Hα and Hβ lines at $\lambda_{\mathrm{rest},\alpha}$ = 6565 Å and $\lambda_{\mathrm{rest},\beta}$ = 4861 Å (Vaporia Astronomy Encyclopedia 2026).

Both of these lines would appear as absorption lines on SN 2026atw's spectrum, although they will not be at their rest wavelengths, but their observed redshifted wavelengths (given that the supernova transient occurs in a host galaxy that is redshifted, or moving away from us). However, pointing out these distinct Balmer lines, or any hydrogen lines alone, alone is indicative of whether SN 2026atw is a Type I or a Type II supernova, which then identifies its progenitor. This is the first step in classifying SN 2026atw.

3.2. *Spectral Analysis: Redshift and Velocity of Ejecta*

After the Hα and Hβ lines have been identified, the next step is to determine at what observed wavelength they are at in SN 2026atw spectrum, as well as calculate the redshift of SN 2026atw by taking the difference of rest from the observed wavelength then dividing by the rest wavelength.

$$z = \frac{\lambda_{\mathrm{observed}} - \lambda_{\mathrm{rest}}}{\lambda_{\mathrm{rest}}} = \frac{\Delta\lambda}{\lambda_{\mathrm{rest}}} \tag{6}$$

After finding z, we can then find the velocity of the supernova ejecta using the following and assuming c = 299,792,458 m/s:

$$v_{\mathrm{ejecta}} = c\,\frac{\lambda_{\mathrm{observed}} - \lambda_{\mathrm{rest}}}{\lambda_{\mathrm{rest}}} = c\,\frac{\Delta\lambda}{\lambda_{\mathrm{rest}}} \tag{7}$$

Given that the velocity is a calculated quantity but redshift is a measured quantity by observing the differences in wavelengths, I wanted to also include the uncertainty with my ejecta velocity calculation. The formula I used for the uncertainty propagation is:

$$\sigma_v = \sqrt{\left(\frac{\partial v}{\partial \lambda_{\mathrm{obs}}}\sigma_{\lambda_{\mathrm{obs}}}\right)^2 + \left(\frac{\partial v}{\partial \lambda_{\mathrm{rest}}}\sigma_{\lambda_{\mathrm{rest}}}\right)^2} \tag{8}$$

where the partial derivatives used are:

$$\frac{\partial v}{\partial \lambda_{\mathrm{obs}}} = \frac{c}{\lambda_{\mathrm{rest}}} \quad \text{and} \quad \frac{\partial v}{\partial \lambda_{\mathrm{rest}}} = -c\,\frac{\lambda_{\mathrm{obs}}}{\lambda_{\mathrm{rest}}^2}$$

These formulae result in the redshift measurement and ejecta velocity with uncertainty calculations for SN 2026atw. The redshift will be compared to the redshift of the host galaxy of SN 2026atw from the DESI legacy survey, which if similar, tells us that our measured redshift is accurate. The velocity of ejecta value will be compared with Figure 1, which allows me to see how my ejecta velocity compares across literature with other Type IIP supernovae.

3.3. *Host Galaxy Redshift, Velocity, and Distance*

The redshift of the host galaxy is given to me from the DESI Legacy Survey database, where $z_{\mathrm{host}} = 0.027$. Given the redshift of the SN 2026atw's host galaxy, I can calculate the the redshift of the host galaxy using c = 299,792,458 m/s with:

$$v_{\mathrm{host}} = c z_{\mathrm{host}} \tag{9}$$

Given that the redshift of the host galaxy is low enough (i.e. z ¡ 1), we can use Hubble's Law to calculate the distance to the host galaxy of SN 2026atw. Hubble's Law is stated as below, where $H_0 = 70$ km/s/Mpc with an uncertainty of $H_{0,\ \mathrm{unc}} = \pm 3$ km/s/Mpc (European Space Agency 2018):

$$v_{\mathrm{host}} = H_0 d \tag{10}$$

where v is the galaxy recession velocity, and d is the distance to the galaxy. Rearranging Hubble's Law to solve for d, given that I'll have calculated the recession velocity of the galaxy, will give me the distance to the host galaxy in Mpc:

$$d = \frac{v_{\mathrm{host}}}{H_0} \tag{11}$$

The uncertainty in the recession velocity (from the H_0 scaling alone) can be calculated as the following given that $H_{0,\ \mathrm{unc}} = \pm 3$ km/s/Mpc:

$$\Delta v_{\mathrm{host}} = v_{\mathrm{host}}\left(\frac{\Delta H_0}{H_0}\right) \tag{12}$$

The velocity uncertainty can be used in finding the uncertainty of calculated distance to the host galaxy using the following formula:

$$\Delta d = \sqrt{\left(\frac{\Delta v_{\mathrm{host}}}{H_0}\right)^2 + \left(d\frac{\Delta H_0}{H_0}\right)^2} \tag{13}$$

Thus, I'll have found the redshift, recession velocity, and distance of the host galaxy for SN 2026atw.

3.4. *Light Curve for SN 2026atw*

The initial light curve data for SN 2026atw that comes from ATLAS forced photometry is given in apparent magnitudes with orange points for those measured in an orange filter and cyan points for those measured in a cyan filter (ATLAS Forced Photometry Service 2026a). The x-axis of the light curve is initially in MJD, which is the Modified Julian Date counting the days from November 17th, 1858 at midnight (Eric W. Weisstein 2026). Although the light curve data is initially in apparent magnitude, I converted these apparent magnitudes into absolute magnitudes because I had the measured redshift of SN 2026atw, that way my light curve was more scientifically significant as to identifying the peak absolute magnitude of the transient. In order to go from apparent to absolute, I used the following established relationship,

$$M = m - \mu \tag{14}$$

where M is absolute magnitude, m is apparent magnitude, and μ is the distance modulus formula seen here:

$$\mu = 5\log_{10}\left(\frac{D_L}{10\,\mathrm{pc}}\right) \tag{15}$$

D_L is the luminosity distance using cosmology, where using Hubble's Law (as consistent with our calculation of the distance to the host galaxy) and the cosmological parameter give me accurate absolute magnitude values with the expansion of the universe. Hubble's constant is defined as $H_0 = 70$ km/s/Mpc and the cosmological constant is defined as $\Omega_m = 0.3$. Together, they help define D_L as:

$$D_L = D_L(z; H_0, \Omega_m) \tag{16}$$

Now that D_L is defined, we can calculate the absolute magnitudes M with the given apparent magnitudes by ATLAS forced photometry as the following with the distance modulus:

$$M = m - 5\log_{10}\left(\frac{D_L}{10\,\mathrm{pc}}\right) \tag{17}$$

Although relevant to knowing the exact date and times of the supernova transient over the light curve, having MJD as our unit of time is not useful in determining the phase, or age, of the transient. I can simply convert my x-axis in MJD to days by using the following formula, where $MJD_{\mathrm{explosion}}$ is an estimate I made in my light curved using a vertical red dashed line based on where my peak magnitude was visually and given supernovae transients can take from days to weeks to reach peak magnitude:

$$t = \mathrm{MJD} - \mathrm{MJD}_{\mathrm{explosion}} \tag{18}$$

Converting my x-axis into days rather than MJD allows me to compare SN 2026atw to other supernovae directly, even with different MJD explosion and MJD peak magnitude dates. This allows me to convert my light curve from brightness vs calendar time to brightness vs explosion phase, which is more relevant in my own light curve analysis of SN 2026atw and comparing to other supernovae. I could also estimate the time it took to achieve peak magnitude using the same formula, but replacing MJD with $\mathrm{MJD}_{\mathrm{peak}}$ to find the t_{rise} that it took to achieve peak magnitude:

$$t_{\mathrm{rise}} = \mathrm{MJD}_{\mathrm{peak}} - \mathrm{MJD}_{\mathrm{explosion}} \tag{19}$$

I would again like to emphasize that $\mathrm{MJD}_{\mathrm{explosion}}$ and $\mathrm{t}_{\mathrm{rise}}$ are estimates and that further modeling of the light curve of SN 2026atw is required in order to get more accurate values for these times.

After doing all of this, the resulting plot should be light curve with absolute magnitude vs time in days of SN 2026atw's transient. By observing the shape of the curve alone, we can identify the subtype of SN 2026atw's classification. Subtyping SN 2026atw not only tells us the kind of light curve we would expect to see from it, but how the progenitor's core-collapse occurred, which produces the shape of the light curve itself.

3.5. *Comparing SN 2026atw to Other Light Curves*

In order to compare my light curve results for SN 2026atw to other supernovae types, I first chose to compare SN 2026atw to a Type Ia supernova, 2017hjw. This supernova was chosen from (Tonry et al. 2018), as it had an excellent, distinctive light curve typical of Type Ia supernovae. Comparing SN 2026atw with other supernovae also allows us to qualitatively understand the differences we would expect to see in light curves between Type I and Type II supernovae.

In order to compare the light curves of SN 2026atw and SN 2017hjw, I produced the lightcurve of SN 2017hjw using the same steps outlined in Section 3.4 knowing from (IAU Transient Name Server 2017) that $z_{2017\mathrm{hjw}} = 0.0161$, which resulted in another absolute magnitude vs time in days light curve for the Type Ia supernova based on the supernova's measured redshift. Then, since the $\mathrm{MJD}_{\mathrm{explosion}}$ dates for the supernovae are different and the light curves occur at different times, I aligned the peak magnitudes of SN 2026atw and SN 2017hjw so that their peak magnitudes were both achieved at t = 0 days on my plot, and both light curves could be compared more accurately, even with the different $\mathrm{MJD}_{\mathrm{explosion}}$ dates for both. Denoted on each curve as well is the peak absolute magnitude and the $\mathrm{MJD}_{\mathrm{explosion}}$ date as well as to emphasize this fitting of light curves.

Much like how plotting the light curve of SN 2026atw to a Type Ia supernovae allows us to qualitatively compare between Type I and Type II supernovae, plotting SN 2026atw with another Type II supernova allows us to see the consistency and comparison of subtypes in light curves across Type II supernovae. The supernova I chose was SN 2023ixf, which is classified as a Type II supernova on the Transient Name Server (Transient Name Server 2023) with a redshift of $z_{2023\mathrm{ixf}} = 0.0008$. Using the same steps of plotting the absolute magnitude vs time in days light curve as SN 2026atw and SN 2017hjw, as well as aligning the peaks as I did in the comparison curves between the two, I produced another comparison plot between two Type II supernovae. I will then use qualitative analysis to further compare and affirm that SN 2026atw is in fact a Type IIP supernova.

3.6. *SNID-SAGE Classification and Comparing SN 2026atw to Literature*

To officially confirm that SN 2026atw is correctly classified as Type IIP, I used SNID-SAGE, supernova classifying software that can be installed locally on one's computer. SNID-SAGE is a supernova spectral classification framework, which compares a supernova spectrum that you provide the software with other known types of supernovae in its database (Stoppa & Smartt 2026).

Upon providing the software a spectra, it performs deterministic pre-processing of the spectra, which normalizes and resamples it into a logarithmic wavelength grid, allowing redshift shifts to be simple translations. Then, FFT-based cross-correlations between the provided spectrum and the database (with 6000 spectra) (Stoppa & Smartt 2026) to identify the best candidate matches across the supernova types and redshifts with the provided spectrum. Rather than just selecting a single, best template, the software ranks mutliple candidates with quality metrics combining cross-correlation strengthm spectral overlap, and clustering methods. This allows for both a more thorough estimate of the measured redshift and supernova type based on best candidates rather than just one.

After, SNID-SAGE then consolidates these results into a classification with associated uncertainty confidence scores, including an image of the spectrum provided and the spectrum it found the best comparison to. I'll be using SNID-SAGE to essentially confirm that SN 2026atw is a Type IIP supernova, as well as compare my calculated redshift and phase to what the software says.

4. **RESULTS / DISCUSSION**

4.1. *SN 2026atw Spectrum and Attributes*

Figure 2 shows the flux-calibrated observed Hiltner 600 spectrum, which is where the sensitivity function that was used to calibrate the flux-calibrated SN 2026atw spectrum in Figure 3 was derived from. As seen in Figure 2, the calibrated spectra and the known, reference flux spectrum of Hiltner 600 are nearly indistinguishable from each other. This is a great sign that flux-calibration of Hiltner 600 was successful, and that our derived sensitivity function will correctly calibrate SN 2026atw's spectrum.

In Figure 3, SN 2026atw's calibrated spectrum can be seen, as well as the Hα and Hβ Balmer lines that I was hoping to find. The presence of these absorption lines alone in the spectrum immediately classified SN 2026atw as a Type II supernova with a massive main sequence star of a mass $> 8M_\odot$ that became a red supergiant before undergoing core collapse as the progenitor, although they do not yet tell me the subtype classification (as this is determined via the light curve). Hα and Hβ are prominent absorption lines in this spectrum, showing clear dips at where they would be located both around their rest wavelength and redshifted wavelengths, allowing them to be easily identified. The peaks of the spectrum at the end of the ends of these Balmer absorption lines are the redshifted, observed wavelengths of Hα and Hβ, which are at $\lambda_{\mathrm{observed},\alpha} = 6739$ Å and $\lambda_{\mathrm{observed},\beta} = 4992$ Å. Using the observed and rest wavelength for Hα in Equation 6, I measured the redshift of SN 2026atw to be z = 0.0265.

Using the Hα rest and observed wavelengths and the speed of light in Equation 7, I calculated the ejecta velocity of SN 2026atw to be $v_{\mathrm{ejecta}} = 7945.76 \pm 102.66$ km/s. Referencing Figure 1 from literature (Gutiérrez et al. 2017), my calculation for the ejecta velocity is in line with the average ejecta velocity across many Type IIP supernovae, as well as within the standard deviation. Despite the uncertainty in my ejecta velocity measurement, it is certain that my measurement is in fact indicative of SN 2026atw being correctly classified as Type IIP; although, officially classifying it as Type IIP requires observing the light curve results in Section 4.3.

4.2. *Host Galaxy Characteristics Results*

Figure 4 from the DESI Legacy Survey (DESI Legacy Imaging Surveys 2026) shows the host galaxy of SN 2026atw, which was searched for using the same RA and Dec coordinates of the supernova at RA 262.2503, Dec 56.7802. DESI DR1 spectra data classifies the host galaxy with a redshift of $z_{\mathrm{host}} = 0.027$, which is quite close to my measured redshift of z = 0.0265 for SN 2026atw. This is great news, as the redshift of both a supernova and its host galaxy should be quite similar, given that both objects would be redshifted at the nearly the same value of z.

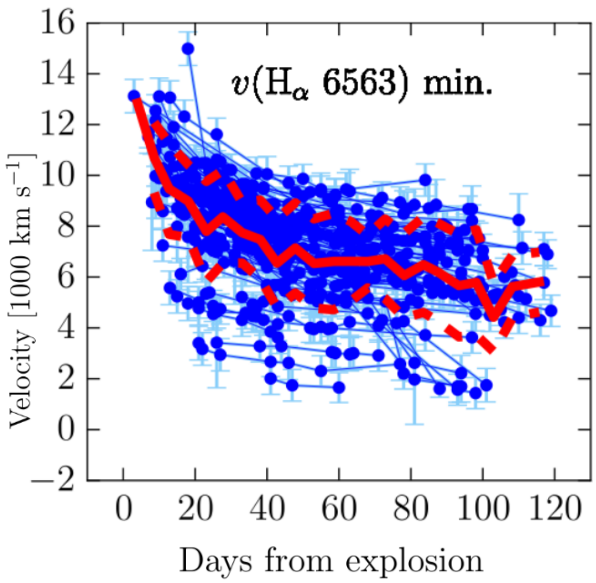

Figure 1. This is a slightly modified figure from Figure 18 of Gutiérrez et al. (2017), where the axis of the full figure were fitted for this specific graph. This figure shows the expansion velocity evolution for Hα. The red solid line is the mean velocity, and the dashed red line is the standard deviation. Qualitatively comparing my calculated velocity of the supernova ejecta at $v_{\mathrm{ejecta}} = 7945.76 \pm 102.66$ km/s at around a phase of 86 $\pm$ 5.0 days, my calculation for the ejecta velocity is consistent within the mean and standard deviation of the Hα line for Type IIP supernovae. This goes to show that my calculations in classifying 2026atw as Type IIP compare well with other Type IIP metrics.

Using $z_{\mathrm{host}} = 0.027$ and the speed of light, I used Equation 9 (which is a valid approximation for low redshift) and found the host galaxy's recession velocity to be $v_{\mathrm{host}} = 8094.40 \pm 346.90$ km/s. With v_{host} and Hubble's constant, I could then use Equation 11 to calculate the distance from us to the host galaxy, which I found to be d = 115.63 $\pm$ 7.01 Mpc. Although the distance to the galaxy and its recession velocity are not necessarily direct classifiers of SN 2026atw being a Type IIP supernova, it's quite neat that we can calculate and estimate these values of the host galaxy just from the spectrum of SN 2026atw.

4.3. *Light Curve of SN 2026atw*

With Methods 3.4 and 3.5, I produced three light curve plots. Figure 5 is the light curve of SN 2026atw with an estimated explosion time at day zero marked by the red dashed line, and with the estimated peak absolute magnitude marked by the green vertical dashed line. In MJD, the estimated explosion day is $\mathrm{MJD}_{\mathrm{explosion}}$ = 61050.59, or day zero as seen on the plot. In MJD, the estimated peak absolute magnitude day is MJDpeak = 61070.59. As seen on the curve and calculated using z = 0.0265 and the apparent magnitude data from ATLAS Forced Photometry on SN 2026atw's light curve, the peak absolute magnitude is $\mathrm{M}_{\mathrm{peak}}$ = -17.65 $\pm$ 0.13. Using Equation 19, I found that $t_{\mathrm{rise}} = 20.0 \pm 5.0$ days, which is how long it took after the explosion for transient SN 2026atw to achieve peak absolute magnitude. Using the data available from the light curve and ATLAS Forced Photometry for SN 2026atw, the phase, or current age of SN 2026atw, is estimated to be 86 $\pm$ 5.0 days.

Most importantly, the shape of the curve tells us the subtype of SN 2026atw, which based on the apparent plateau of the points on the light curve, I classified SN 2026atw as a Type IIP supernova, where the "P" stands for plateau. This plateau is a feature commonly seen in Type II supernovae, as it is caused by the photosphere's recession through the hydrogen envelope progenitor (a massive main sequence star of a mass $> 8M_{\odot}$) as the temperature remains essentially constant during core-collapse (Swinburne University of Technology 2026b). As the supernova expands and cools during the core-collapse of the progenitor, the the photosphere recedes to the star's center and through the hydrogen envelope; it is the depth of this hydrogen envelope that determines the length of the plateau of the light curve. Although the photosphere is receding, the star continues to expand, which forces the photosphere in deeper until the successive regions cool to a temperature cold enough for recombination after the supernova. The plateaus seen in SN 2026atw's light curve, as well as those of other Type IIP supernovae, are essentially due to the action of the receding photosphere in the progenitor.

4.4. *SN 2026atw vs SN 2017hjw Light Curve*

Now that I have classified SN 2026atw as a Type IIP supernova, I can compare it with other light curves of other supernovae. For this first comparison, I chose SN 2017hjw, a known Type Ia supernovae with a distinct light curve typical of Type Ia supernovae. Figure 6 shows the light curve of 2017hjw in red, where the peaks of both supernovae are aligned as to see the change in magnitude over time.

As one notices, in the light curve of SN 2017hjw, the plateau feature seen in SN 2026atw's curve is simply not there, as the progenitor of SN 2017hjw is not a massive star, but a white dwarf that reached the Chandrasekhar limit of $\approx 1.44\,M_{\odot}$. This means that the recession of the photosphere of the progenitor that creates the plateau feature in the Type IIP light curve will never be present in the light curve of a Type Ia supernova, given that their progenitors are white dwarfs that undergo ther-

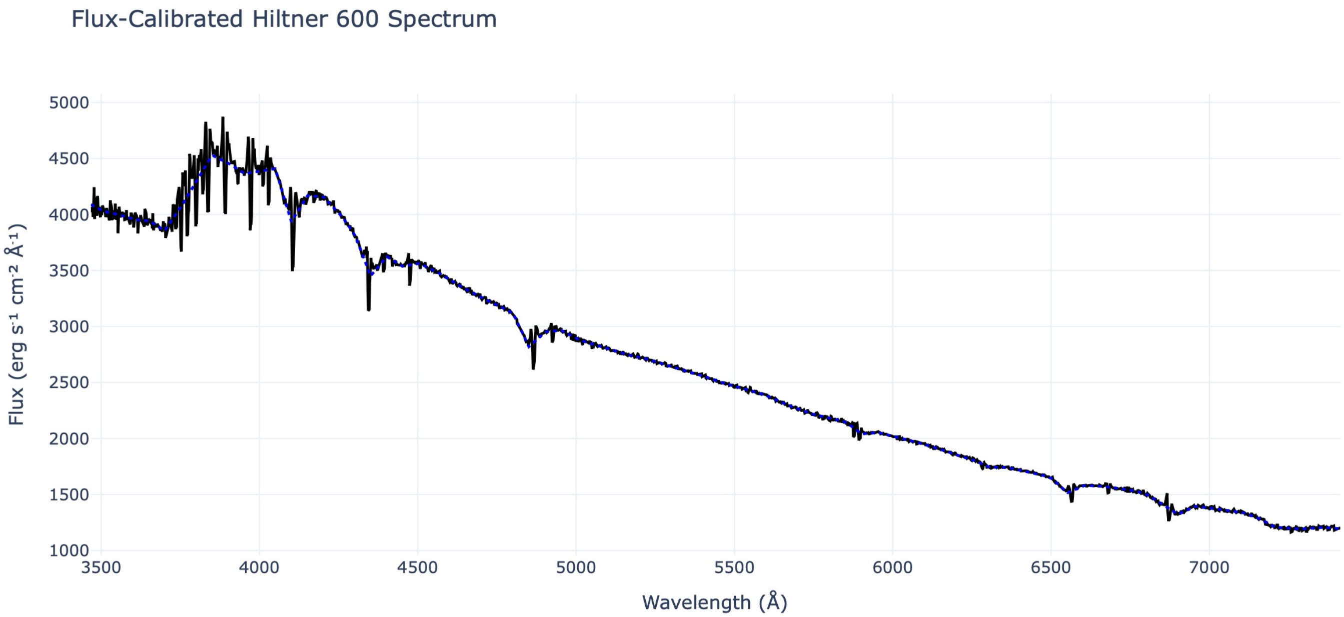

Figure 2. Flux-calibrated spectrum for Hiltner 600. This was made using file 0053.Hiltner600.fits and flux calibrating it with fhilt600.dat.txt. The black line is the flux calibrated spectra, while the dotted blue line is the reference flux spectra from fhilt600.dat.txt.

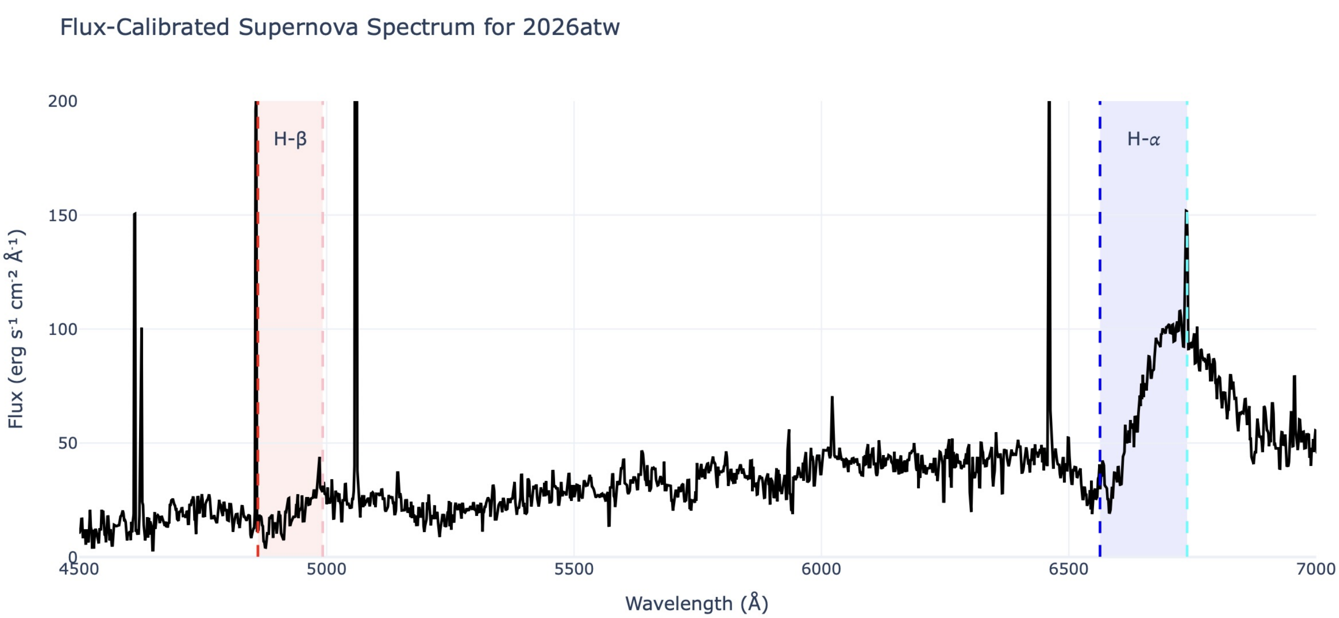

Figure 3. Flux-calibrated spectrum for 2026atw. This was made using file 0080.2026atw.fits and flux calibrating it with the sensitivity function derived from calibrating Hiltner 600's spectrum. The black line is the calibrated spectrum of 2026atw. The red dashed line is the rest wavelength for Hβ at $\lambda_{\mathrm{rest},\beta} = 4861$ Å, and the light pink dashed line is the redshifted, observed wavelength for Hβ at $\lambda_{\mathrm{observed},\beta} = 4992$ Å. The dark blue dashed line is the rest wavelength for Hα at $\lambda_{\mathrm{rest},\alpha} = 6565$ Å, and the cyan dashed line is the redshifted, observed wavelength for Hα at $\lambda_{\mathrm{observed},\alpha} = 6739$ Å. $\Delta\lambda$ for Hβ is shaded red, while the $\Delta\lambda$ for Hα is shaded blue. The calculated redshift of 2026atw from these hydrogen lines is z = 0.0265.

Figure 4. This is a screenshot of 2026atw's host galaxy circled in white at RA 262.2503, Dec 56.7802 in the DESI Legacy Survey (DESI Legacy Imaging Surveys 2026). As highlighted in the screenshot, the redshift of the host galaxy is 0.027, which is consistent with both mine and SNID-SAGE's calculated redshifts of 2026atw of z = 0.02687 and my calculated redshift of z = 0.0265.

monuclear fusion rather than core-collapse stars with a hydrogen envelope.

It makes sense to compare the light curve of SN 2026atw to a Type Ia supernova as Type Ia supernova have consistent light curves given that their progenitors explode with nearly uniform, consistent peak brightness, making Type Ia supernovae good standard candles for measuring distances and observing the dark energy accelerated expansion of the universe (Lawrence Berkeley National Laboratory 2014).

4.5. *SN 2026atw vs SN 2023ixf Light Curve*

Since I have compared SN 2026atw to a Type Ia supernova, SN 2017hjw, I also want to compare it to another Type II supernova as to see similarities rather than differences in the light curves that further affirm SN 2026atw's classification as Type IIP. The other Type IIP supernova that I chose was 2023ixf, which is classified as a Type II supernova whose light curve data was also available on ATLAS forced photometry.

Comparing the light curves of SN 2026atw and SN 2023ixf, one sees many more similarities between these two light curves than that of SN 2026atw to SN 2017hjw. Both light curves feature a plateau feature, although 2023ixf isn't classified as Type IIP on the Transient Name Server, but rather as Type II. This is due to how there is much more variabiltiy in Type II supernovae light curves than Type Ia light curves, given that Type II supernovae don't all reach the same known peak magnitude and are different based on the size and mass of the progenitor.

Chances are that although SN 2023ixf and SN 2026atw both have massive star progenitors, their progenitors vary in both mass and size, thus causing their light curves to achieve different peak magnitudes. How big the value of the peak magnitude is significantly dependent on the total energy of the supernova explosion (Carroll 2001). This total energy of the explosion is dependent on the size of progenitor, as generally, bigger progenitor stars mean bigger explosion and overall greater total energy.

However, in order to determine the mass, size, and explosion energy of SN 2026atw with certainty, further modeling of my light curve to other Type IIP curves is required to do so and therefore not covered in this paper. This modeling can be done in the future by using hydrodynamic modeling of the event or using pre-explosion observations, which are done on the light curves and ejecta velocity data (Utrobin & Chugai 2008). Another example of a modeling method used to determine the mass and size of progenitor is by age-dating, which can be done by analyzing the stellar cluster where the supernova exploded and approximating the age and mass of stars within it. This is all work left for the future in further classifying SN 2026atw and its progenitor properties.

4.6. *SNID-SAGE and Official Classification*

Using the SNID-SAGE software, I uploaded the spectrum of SN 2026atw so that it would also be classified via software rather than just analytical methods. SNID-SAGE found that the best candidate comparison supernova to SN 2026atw was Type IIP supernova sn1992H, which is represented by the much smoother, orange spectrum seen in Figure 8.

Some important features to notice in Figure 8 as we compare the two spectra are how the Balmer lines, specifically Hα and Hβ, in SN 2026atw seem to follow the same curvature of sn1992H's spectrum. Overall, SN 2026atw's spectrum compares nicely to the emission and absorption lines in sn1992H's spectrum, showing how having the same spectral features between supernovae can lead to classification (especially by identifying prominent spectral lines like Hα and Hβ).

SNID-SAGE estimated the redshift of SN 2026atw to be z = 0.02687 $\pm$ 0.0057, which is very close to my calculated redshift of z = 0.0265. Comparing these two redshifts to that of the host galaxy from the DESI Legacy Survey at zhost = 0.027, it is a great sign that

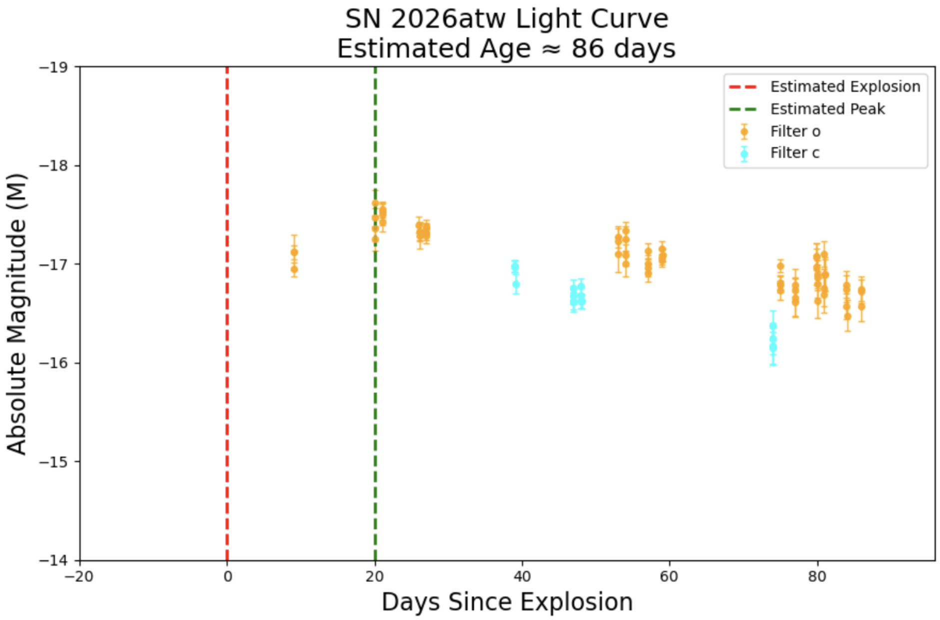

Figure 5. This is the light curve of 2026atw using the supernova apparent magnitude data from ATLAS Forced Photometry (ATLAS Forced Photometry Service 2026a). Given that the redshift of 2026atw was calculated at z = 0.0265, the absolute magnitude light curve was able to be produced using the distance modulus formula. The red dashed line is just an estimate of the explosion date, which usually occurs a few weeks before the peak magnitude of the supernova (further modeling of this light curve with other Type IIP curves is required for a more accurate explosion date). The green dashed line in this curve marks the peak magnitude date, which is at $t_{\rm rise}$ = 20.0 ± 5.0 days at $M_{\rm peak}$ = -17.65 ± 0.13. Using the estimated peak and explosion date, the phase of the transient is estimated to be around 86 ± 5.0 days past the explosion date.

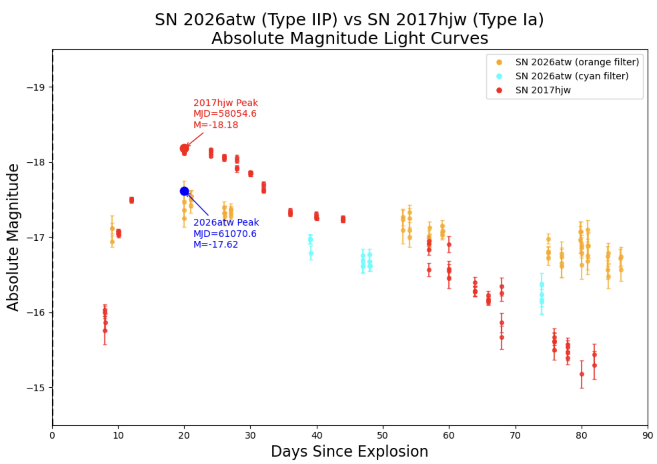

Figure 6. This figure shows a comparison of 2026atw's light curve, which was classified as a Type IIP supernova, to the light curve (in red) of 2017hjw, a known Type Ia supernova. The light curve data for 2017hjw was found in Figure 12 (Tonry et al. 2018, pg 17), where querying the ATLAS forced photometry database gave the light curve data of 2017hjw (ATLAS Forced Photometry Service 2026b). The peaks of both curves are aligned as to better compare side by side the shapes of the curves from the peak magnitude.

all three redshifts are within accuracy of each other (as they should be, given that SN 2026atw lives in the host galaxy and should have a redshift similar to it).

SNID-SAGE calculated the phase or age of SN 2026atw to be 103 days, which is a couple of days off from my estimated value of 86 ± 5.0 days from my SN 2026atw light curve. This difference is to be expected, as my value for the phase is based on estimation of the explosion date of SN 2026atw and would require further modeling and fitting in order to get an accurate explosion date and thus a more accurate phase. Overall, my methodology in classifying SN 2026atw as Type IIP is quite consistent with SNID-SAGE's classification and comparisons of characteristics and properties of SN 2026atw to other Type IIP supernovae.

5. CONCLUSION

My goal for my Astronomy 100 final project was to observe and classify a supernova transient. The supernova transient I chose was SN 2026atw, a transient visible from the FLWO in Arizona that was previously unclassified until now, where it's published classification can be seen on the Transient Name Server for 2026atw. Using the data and methods outlined in this paper, I classified SN 2026atw as a Type IIP supernova and its host galaxy with the following characteristics:

- SN 2026atw is a Type IIP supernova, where "P" stands for the plateau feature seen in its light curve. Its classification of Type II was deter-

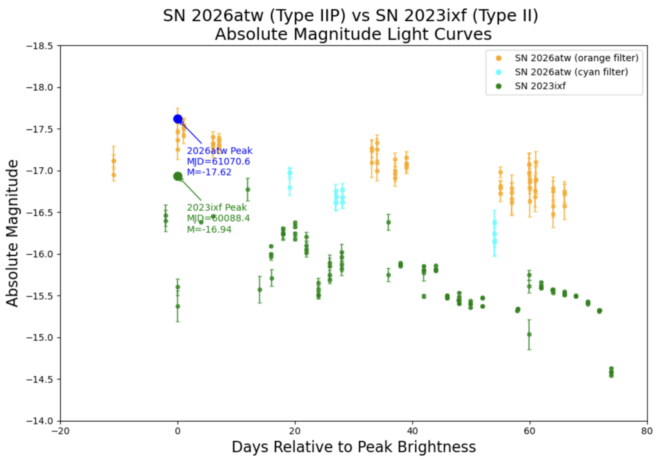

Figure 7. This figure shows a comparison of 2026atw's light curve, which was classified as a Type IIP supernova, to the light curve of 2023ixf, classified as a Type II supernova. The light curve data for 2023ixf was found on the Transient Name Server (Transient Name Server 2023), where querying the ATLAS forced photometry database gave the light curve data of 2023ixf (ATLAS Forced Photometry Service 2026c). The peaks of both curves are aligned as to better compare side by side the shapes of the curves from the peak magnitude. Its important to note that the light curves for Type II supernovae are not as consistent as those of Type Ia supernovae; however, qualitatively, comparisons can be drawn between 2026atw and 2023ixf that further verify 2026atw's classification as Type IIP.

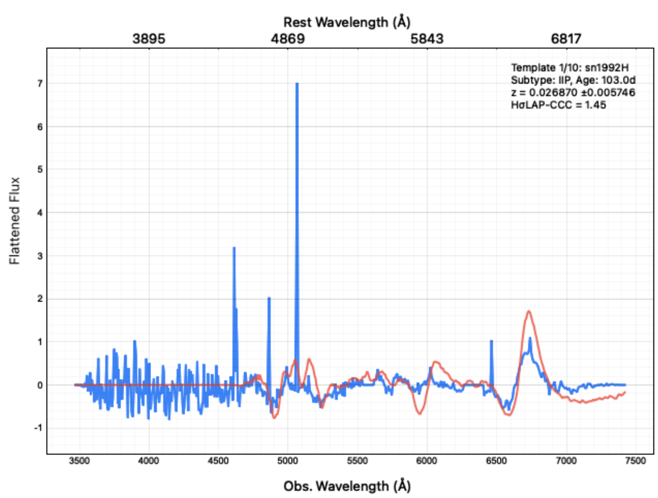

Figure 8. From the SNID-SAGE software, this is the spectrum of 2026atw (in blue) being compared with another Type IIP supernova, sn1992H (in orange). The software's estimate for the phase of the transient is 100+ days, and it's calculated redshift is consistent with the it's host galaxy's redshift as classified in the DESI Legacy Survey (DESI Legacy Imaging Surveys 2026)

mined by the presence of Balmer lines in its flux-calibrated spectrum.

- The progenitor of SN 2026atw is a massive main sequence star with a mass $> 8M_\odot$ that became a red supergiant and underwent core collapse. Further modeling of the light curve of SN 2026atw is required to determine the size and mass of the progenitor with certainty.
- The redshift of SN 2026atw was calculated to be z = 0.0265 from my own redshift measurement of the flux-calibrated spectrum. This is consistent with its host galaxy's redshift of $z_{host} = 0.027$ and SNID-SAGE's calculated redshift of z = 0.02687.
- The ejecta velocity of SN 2026atw is $v_{ejecta} = 7945.76 \pm 102.66$ km/s. This is consistent with the literature of calculated ejecta velocity at its phase of 86 ± 5 days using the Hα line as seen in Figure 1.
- Using Hubble's Law, the host galaxy recession velocity was calculated to be $v_{host} = 8094.40 \pm 346.90$ km/s, and its distance from us was calculated to be d = 115.63 ± 7.01 Mpc.
- From the light curve of SN 2026atw, the peak absolute magnitude M is around $M_{peak} = 17.65 \pm 0.13$. This peak magnitude was estimated to be achieved at $t_{rise} = 20.0 \pm 5.0$ days after the estimated explosion date. The total explosion energy of the supernova that contributed to this peak absolute magnitude requires more modeling of the light curve to be determined.
- The estimated phase, or age, of SN 2026atw from the light curve is around 86 ± 5 days. This is similar to SNID-SAGE's phase of 103 days, as well as consistent with Figure 1 and the ejecta velocity measurement of $v_{ejecta} = 7945.76 \pm 102.66$ km/s.

5.1. *GitHub Repository*

The link to the GitHub repository with all of the code used in this project can be found here: Astro100-final-project.

5.2. *Acknowledgments*

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